

MUHAMMAD RAFAY IRFAN

Software Engineer · Python · C++ · C · AWS · End-to-End Product Delivery

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Work Experience

Google Summer of Code (lib2geom, Inkscape)

May 2025 – Sep. 2025

Open Source Developer

Remote

- Owned a complex deliverable end-to-end, scoped the problem from first principles, selected numerical methods, implemented in C++, and shipped with a 50+ case Google Test suite covering tangent, overlapping, and precision-sensitive edge cases, merged into a production codebase used by millions.
- Wrote high-quality, maintainable C++ to resolve a long-standing robustness gap in lib2geom's intersection engine, replacing fragile floating-point logic with error-bounded interval arithmetic, directly improving correctness for degenerate curve-curve configurations.
- Provided and incorporated constructive code review direction across async feedback cycles with senior open-source maintainers, raising the bar on API surface design and edge-case handling to meet the standards of a mature geometry library.

Inkscape

Oct. 2024 – May 2025

Open Source Developer

Remote

- Independently clarified ambiguous technical requirements by reading unfamiliar codebases and writing minimal reproducible test cases, diagnosed and resolved high-impact UI and path-manipulation bugs across rendering, tool interaction, and editor state layers in a multi-hundred-thousand line C++/GTK project with no onboarding documentation.
- Drove continuous quality improvements for 3M+ active users with targeted, regression-free fixes, maintained CI integrity across 5+ dependent tool behaviors, and communicated technical decisions clearly in async review threads without direct mentorship.

Systems Limited

Jul. 2023 – Sep. 2023 & Jul. 2024 – Sep. 2024

Software Engineering Intern

Lahore, Pakistan

- Designed and deployed cloud solutions on AWS (EC2, S3, RDS, Lambda) and strengthened security across 3 production environments via IAM policies, bucket access controls, and serverless event-driven functions.
- Worked under the Cloud App Development team, gaining hands-on training in Java and contributing to enterprise application development practices.

Projects

Real-Time Healthcare AI Voice Agent | *Node.js, TypeScript, Twilio, AWS Bedrock, ECS/Fargate, PostgreSQL, S3*

- Shipped a production-quality, HIPAA-eligible real-time voice system integrating Twilio Media Streams with AWS Bedrock Nova Sonic, with end-to-end ownership from architecture through production rollout, including KMS-encrypted S3 pipeline, PostgreSQL audit trails, and a full integration test suite.
- Config-driven multi-agent design (Node.js, TypeScript) dynamically loads each agent's prompt, voice, locale, tool permissions, and call rules per call, supporting 20 distinct agents without code duplication, maximizing reusability and maintainability.

SmartEngage | *Python, FastAPI, XGBoost, PyTorch, LangChain, PostgreSQL, Docker, ChromaDB*

- Built end-to-end in Python/FastAPI with modular, reusable service layers: ingests social media telemetry via async pipelines, predicts content virality with a hybrid XGBoost + transformer model, and routes underperforming drafts through an agentic RAG loop backed by a vector DB and LLM orchestrator, all containerized with Docker for consistent deployment.

Blood Bank Management System | *PHP, MySQL, PDO, JavaScript, HTML5, CSS3*

- Designed a normalized relational schema across 7 entities and built a full-stack PHP/MySQL platform enabling real-time blood inventory tracking and donor-to-center matching; implemented role-based access control, PDO-parameterized queries to eliminate SQL injection, and separate admin and donor portals with live inventory management and medical record access.

Education

Ghulam Ishaq Khan Institute

2022 – 2026

BS. Computer Science

Skills

Languages: C++, Python, TypeScript, JavaScript, Java, C **Frontend:** React, Next.js, Tailwind CSS

Backend & APIs: FastAPI, Node.js, REST API Design, Docker, Kubernetes (familiar) **Databases:** PostgreSQL, MySQL, ChromaDB

ML / AI: PyTorch, XGBoost, Scikit-learn, Transformers, RAG, LangChain, LLMs, Pandas, NumPy

Cloud & Infra: AWS (EC2, S3, RDS, Lambda, ECS/Fargate, Bedrock) **Tools:** Git, CMake, Google Test, Linux